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S.101 SOIL INVESTIGATION

1. SCOPE OF WORK

The scope of Soil Investigation shall be such as to clarify the soil conditions in order that satisfactory and economical foundations can be designed and proposed recommendations for suitable types of foundation and establishing the reliable soil parameter based on the analysis of the geotechnical data.

These investigations shall be performed immediately after the Contract Award. The Bidder shall submit the investigations program to the Owner for approval. The approved program shall not relieve the Bidder from his obligation to undertake additional investigations, if required by the Owner after evaluation on the first report for the soil investigation. The cost for the additional investigation shall be included in the Contract Price.

Investigation and elaboration of the report on investigation results and structures foundations conditions shall be entrusted to an institution specialized for soil mechanics and foundation. Such an institution and the report shall be approved by the Owner.

The work shall include, but not limited to, the mobilization and demobilization, transports and furnishing of all machines, equipment and installations necessary to perform the works specified hereafter and not limited to, such as drilling rigs, rods, bits, casings, CPT, power generators, compressors, pumps, vehicles, cisterns and all other matters deemed necessary by the Bidder to perform the foreseen investigations as indicated in Part 4, Subsection 1.2 “Summary of Work” and Technical Specification S-001 “General Design Requirement of Structure, Civil and Building Works”.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”. The Codes and Standards of the following organizations extracted from the Part 4, Section 2, Subsection 2.5 “Codes and Standards” are specifically applicable to the design, manufacture and testing of the work included in this Specification.

ASTM	:	American Society for Testing and Materials
JIS	:	Japanese Industrial Standards
BS	:	British Standard
AASHTO	:	American Association of State Highway & Transportation Officials.

3. WORKS EXECUTION

3.1 SCOPE OF INVESTIGATION

3.1.1 The following investigations program shall be minimum and compulsory:

- 1) Sampling and coring
- 2) Standard penetration test
- 3) DCPT (Dutch Cone Penetration Test)
- 4) Physical property test
 - a) Moisture content, Wet Density, Water Content
 - b) Specific gravity
 - c) Sieve analysis and Hydrometer Analysis
 - d) Liquid limit, plastic limit, Plasticity index, Shrinkage Limit
 - e) Void Ratio
 - f) Degree of Saturation
 - g) Chemical analysis of soil
- 5) Mechanical property test
 - a) Unconfined compression test
 - b) Direct shear test
 - c) Consolidation test
 - d) Permeability test
 - e) Triaxial Comp. (UU Test)
 - f) Triaxial Comp. (CU test)
- 6) Meteorological survey and Seismic study
- 7) Geo-electrical
- 8) Test of seawater quality
- 9) Hydro-oceanografi survey

3.1.2 Quantity Test

The following minimum quantity of boring, sampling, SPT test and frequency of laboratory shall be test compulsory:

- 1) The boring holes shall be arranged at each structural area as following, STG (Steam Turbine Generator) and Central control building area, HRSG area, Water intake area, Discharge area and other area approved by the Owner in the Project area.
The borehole data location provided by the Owner shall be considered by the Bidder.
- 2) The boring shall be confirmed to the depth of designed soil bearing layer with N-value not less than 50, this value shall be the same 3 times respectively and confirmed that such layer should have not less than 5 m depth.
- 3) The boring holes shall be expected to the depth of 50m approximately.
- 4) Static cone penetrations test on shore to the depth with $q_c = 250 \text{ kg/cm}^2$.
- 5) SPT and sampling shall be performed in 2.0m intervals.
- 6) Undisturbed sample of soil for laboratory test shall be taken minimum 1(one) at each different layer which quantity shall be 15% (percent) of the total number of boring holes or more.
- 7) The laboratory test shall be performed minimum 1(one) for each layer those which quantity shall be at 15% (percent) of the total number of boring holes.
- 8) Cone Penetration Test: This test shall be carried out in accordance with ASTM D3441 which is sometime referred to as the “Dutch Cone Test”.
- 9) Resistivity logging: Resistivity logging shall be conducted to determine electrical resistance of soils at various depths for design of the ground electricity. It needs minimum 2 (two) holes and to find out AQUIFIER Potential / Water Resources.
- 10) Plate loading test: This test shall be performed in accordance with ASTM 1195, at the level of 1m depth below the ground level, minimum 2 holes.

- 11) Permeability test: To determine the permeability of subsoil, field permeability test shall be conducted, minimum 2 holes.
- 12) Chemical check of soil and underground water: The purpose of chemical analysis is to study influence of soil and groundwater on buried cement and steel. It needs minimum (2) holes.

4. CORE DRILLING

4.1 PROCEDURE

- 4.1.1 Core drilling shall be carried out by rotational method with clear water flush or on the request of the Owner without any flush at all.
- 4.1.2 Core recovery shall be required on the entire length of all drillings. A minimum core recovery of 90% is requested. Only material not altered or damaged by the drilling process will be considered as “core”. Zones where SPT, vane tests or undisturbed sampling was performed, are considered as zones with recovered core, every layer shall be “Vane Test” with minimum 2 holes.
- 4.1.3 Cores have to be handled carefully, in order to prevent deformation or breaking.

4.2 DEPTH

- 4.2.1 Drilling equipment must be able to perform bore holes to a minimum depth of 50m.
- 4.2.2 The final depth of the bore hole, as well as the depth achieved with every core run will be measured with the drill rods in accuracy of 5 cm.
- 4.2.3 If a boring does not reach the required depth, an additional hole has to be drilled and no compensation will be given for the abandoned hole.
- 4.2.4 The Owner is entitled to change the length of borings based on geological features.

4.3 DIAMETERS

4.3.1 The initial diameter will have to be chosen by the Bidder. The final core diameter should not be less than 54 mm.

4.3.2 In order to allow proper telescoping, core barrels, bits and casing for several diameters should be on the site.

4.4 OTHER OBLIGATIONS TO THE BIDDER

4.4.1 The Bidder shall assist the Owner during core logging and for all other work related to the preparation and supervision of the Bidder's work.

4.4.2 The Bidder shall take colored picture of all cores. Two copies of 9 x 13 cm in size and the negative shall be handed over to the Owner.

4.4.3 Each finished bore hole shall be marked by a concrete block cast in place, with a minimum base dimension of 40 x 40 cm and a minimum height above ground surface of 10 cm. The number of the bore holes shall be engraved in the concrete.

4.5 EQUIPMENT

4.5.1 Rigs shall meet the following specifications:

- Hydraulic or mechanical feed
- Transmission gear to adjust spindle speed (e.g. 60 to 300 rpm).

4.5.2 The core barrels shall be in perfect, which guarantee the request core recovery. The use of single - tube core barrels is not allowed.

4.5.3 Diamond bits and hardened steel bits in perfect condition from a well-known manufacture shall be used.

4.5.4 The Bidder shall supply all necessary flush water for the execution of boring and pertinent tests. Facilities for sedimentation and water reservoir of adequate capacity may be necessary. Only clear water shall be used.

4.5.5 The Bidder shall establish the sequence of drilling and testing based on the frequency requirements for the different bore hole tests to guarantee testing

and sampling in natural soil conditions and to allow for a reasonable drilling progress. The sequence shall be approved by the Owner.

- 4.5.6 Core boxes should be made of good and durable material and covered by a lid with hinges.

The inner length of a core box shall be 1 m and for each meter of bore hole one meter of core box shall be foreseen. The end of each core, as well as other pertinent information (e.g. core loss, extracted sample, etc) has to be made immediately in the core boxes with indication of actual depth.

Inscriptions on the boxes have to be punched into aluminum or plastic straps. The boxes must show on the inside of the lid the following indications:

- Job site
- Number of bore hole
- Date of start and finishing the bore hole
- Number of the box
- Depth interval

The boxes must show on the outer small side the following indications:

- Job site
- Number of bore holes
- Number of the boxes and some of boxes (for example = box number 5 from 6)

Core boxes have to be stored in such a way that they are protected against rain and sun.

5. STANDARD PENETRATION TEST (SPT)

- 5.1. The test shall be made according to the following standards: ASTM Designation D 1586 or BS 1377, Test 19.
- 5.2. The extracted sample shall be packed and sealed in a plastic bag and placed in the core box. It shall be labeled for identification with date, bore hole number, depth and number of blows.
- 5.3. With the sampler resting on the bottom of the hole, drive the sampler with blows from the 140 pound hammer falling 75 cm until either 45 cm have

been penetrated or refusal (defined as 10 blows per 1,5 cm or less of penetration) has been reached.

- 5.4. Standard reporting shall give the number of blows for the first, second and third penetration of 15 cm. If the number of blows for the first 15 cm would exceed 20, the penetration depth for the first, second and third 20 blows shall be reported. Further more it shall be reported if the hammer is free chopping or fixed by a rope.

6. VANE TEST

- 6.1. The test shall be done according to the standard ASTM D 2573, AASHTO T223, BS 1 377, T1 8 or another standard approved by the Owner.
- 6.2. To allow for calibration it shall be performed just below the location of a SPT test.
- 6.3. The test shall be performed in such a way that the rods are isolated from the soil, i.e. that no friction occurs between the rods and the soil.
- 6.4. The maximum and residual value of torque and corresponding shear strength shall be reported.

7. PERMEABILITY TEST

- 7.1. If there is any sandy layers of the subsoil, the permeability shall be measured with the falling head test method.
- 7.2. The test shall be performed where such a layer is encountered or as directed by the Owner.

In general not more than 2 tests shall be performed in one sand layer and no test shall be closer than 3 m to the next one.

- 7.3. Before performing the test, the bottom of the bore hole shall be cleaned and the water within the casing shall be clear. The ground-water level at the time of the test shall be registered.

- 7.4. For the test, the casing shall be filled with clear water and the falling of the head in function of time shall be measured. Registration time may be up to 20 minutes.
- 7.5. After the test, the depth of the bore holes has to be controlled.
- 7.6. If the casing is telescoped, special measures are needed to avoid water loss at the level of diameter change.
- 7.7. The record of tests shall include:
- depth of boring
 - depth of ground-water level
 - duration of test
 - diameter of casing at depth of casing and water level measurements
 - array of data with time and fall of water level during test.
- 7.8. The Bidder shall interpret the data and give the permeability of the measured soil layer.
- 7.9. Data and interpretation shall be handed over to the Owner on approved forms within 48 hours.
- 7.10. The compensation for test is independent of the diameter of the depth below surface where the tests were performed.
- 7.11. The supply of the required water volume for flushing the hole and testing is understood to be part of the test and the Bidder is not entitled to any additional payment irrespective of the volume of water lost in preparation and performance of the test.

8. SAMPLING

- 8.1. In the course of the field investigations, 3 types of samples shall be distinguished:
- undisturbed soil samples
 - disturbed samples
 - SPT-sample

All samples shall be taken from the drillings. Within counting the SPT-samples, one sample shall be taken every in 2 m interval

- 8.2. Undisturbed soil samples are extracted from bore holes with a special instrument, as defined in ASTM D 1 587 in finegrained or cohesive soils. The instrument shall be subject to the Owner's approval. Undisturbed soil samples shall be taken in appropriate soils every 5 m interval or at regular intervals, as directed by the Owner.
- 8.3. Disturbed samples shall be of soil, taken from cores and packed and sealed in plastic bags or plastic tubes. Number, type and size of these samples shall be directed by the Owner.
- 8.4. All samples shall be labeled. The label shall indicate the Project, Site, bore hole, depth and date of extraction.
- 8.5. On the Site, the samples shall be stored in a special place sheltered from sun and rain.

9. CONE PENETRATION TEST (CPT)

- 9.1. CPT equipment and testing procedure shall be in accordance with the recommended standard
Deviations from that standard shall be indicated in the test report and need to be approved by the Owner.
- 9.2. Based on the actual geological knowledge depth are envisaged for testing 30 m.
- 9.3. The test results shall be presented in a plot of cone resistance and total side friction versus depth.
- 9.4. Additionally but not alternatively the Bidder may offer the measurement of the pore water pressure versus depth. In this case the Bidder shall submit a description of this equipment and indicate the price for such readings per linear meter as an over item to the price of CPT.

- 9.5. The test records shall be handed over to the Owner within 48 hours after test performance.
They shall also be integrated in the final report.

- 9.6. The cost for setting up of the test equipment shall be included in the Contract Price.

10. LABORATORY TESTING

- 10.1. The samples taken in the field shall be transported to the laboratory at least every fortnight.

- 10.2. The Bidder shall be responsible for adequate storage, extraction, description and preparation of each sample. All this work shall be included in the Contract Price and no extra compensation is given for such work.

- 10.3. The Bidder will receive, from the Owner, a test list indicating the tests to be performed for each sample. If, for any reason, this test program can not be followed, the Bidder shall immediately inform the Owner and furnish reasons.

- 10.4. The number of each test given is the minimum requirement. Variations from the given quantities may be directed by the Owner.

- 10.5. For each test, the Bidder shall indicate the standard he will follow.

- 10.6. Deviations from any standard or the use of other standards than the ones specified hereafter, need the written approval of the Owner.

- 10.7. The following tests for soil properties shall be performed in accordance with ASTM and/or AASHTO and/or BS standards. Test and data recording shall be included.

- 10.1.1 Particle-size analysis of soil, ASTM D422 or AASHTO T88 or BS 1377.

- 10.1.2 Liquid limit of soil, ASTM D423 or AASHTO T89 or BS 1377.

- 10.1.3 Natural water content, ASTM D2216 or BS 1377.

- 10.1.4 Natural (moist) unit weight, BS 1377.
- 10.1.5 Specific gravity, ASTM D854 or AASHTO T100 or BS 1377.
- 10.1.6 Organic matter, ASTM C40 or ASTM T21 or BS 1377.
- 10.1.7 Permeability, ASTM D2434
- 10.1.8 Unconfined compression strength, ASTM D2166 or AASHTO T208 or BS 1377.
- 10.1.9 Consolidation, ASTM D2435 or ASHTO T216 or BS 1377
- 10.1.10 Triaxial shear strength, AASHTO T266 or BS 1377
- 10.1.11 Direct shear strength, AASHTO T236

11. REPORTS

11.1 DAILY REPORTS ON FIELD WORK

- 11.1.1 The driller will daily carry out a report, containing:
 - diameter and type of bit
 - diameter and depth of casing
 - rotation velocity
 - drilling pressure
 - total water losses
 - pressure of clear-water flush, color of flush
 - depth and time interval of core runs
 - types of soil
 - water level in hole at beginning and at end of shift, intermediate measurements
 - any other pertinent information
- 11.1.2 The form of the daily report shall be subject to the Owner's approval.
- 11.1.3 There is no extra compensation for the daily reports.

11.2 FINAL REPORT ON FIELD WORK

The final report shall incorporate the following items:

- 11.2.1 Activity time schedule of investigation program (table)
- 11.2.2 Layout of the area where Muara Towar Gas Fired CCPP is to be constructed, with structures locations indicated, and the positions of investigations carried out.
- 11.2.3 Drilling equipment used by the Bidder on site (list)
- 11.2.4 Comments on the performed work.
- 11.2.5 Drawings of the bore logs.
- 11.2.6 Complete data and interpretation for all field tests as SPT and permeability and so on.
- 11.2.7 List of all extracted samples (SPT, disturbed and undisturbed)
- 11.2.8 Logs of CPT.

11.3 DRAFT REPORT

A draft of the final report shall be subject to the Owner's approval. It shall be delivered not later than 1 week after termination of the field work.

11.4 PROGRESS REPORT OF LABORATORY WORK (INCLUDING RECOMMENDATION PILE TYPE AND SETTLEMENT ANALYSIS)

Every two weeks, the Bidder shall prepare and forward to the Owner one report with the following features:

- Samples received
- Tests performed
- Deviation from the test program
- Comments on the test program and on the test result
- An annex containing all test results of the given period

No extra compensation is foreseen for the Progress Reports.

11.5 TEST RECORD

All forms for the test readings and interpretations shall be approved in advance by the Owner. The Owner may also require additional forms or different plots.

11.6 FINAL REPORT ON LABORATORY WORK

11.6.1 The final report shall contain the following items:

- a sample list
- a list of performed tests
- summaries of test results, arranged according to technical, geological or geotechnical features
- all of test results, readings, plots and interpretations.

11.6.2 The Bidder shall first prepare a draft issue of the final report and submit it to the Owner for approval. It shall be delivered not later than 4 weeks after termination of the field works.

11.6.3 All form and reports shall be written in English and Indonesian language.

11.6.4 The Bidder shall hand over 5 copies of the final reports with softcopy (pdf) to the Owner. Furthermore, the Bidder shall provide to the Owner, the original of all text and data sheets, and a transparent or softcopy (pdf) of all drawings.

11.6.5 The cost for reporting shall be lump sum and shall include all data collection, typing, copying, interpretation, calculation, designing, drawings, reproducing, posting, as well as all paper and other material and work necessary, to provide the required and approved reports.

11.6.6 The works required in this Specification shall be regarded to be terminated after the Owner has accepted in writing, the draft of the final reports and after the Bidder has delivered all originals and copies of the final reports in correspondence with the accepted draft reports, as specified above.

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